

Related Rates

ANSWER KEY



Setting up related-rates problems

- 1 A circular oil spill is expanding so that its radius increases at 3 m/min. How fast is the area increasing when the radius is 50 m?
 $A = \pi r^2$, so $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2\pi(50)(3) = 300\pi \approx 942.5 \text{ m}^2/\text{min}$.
- 2 A 10 m ladder leans against a wall. The bottom slides away from the wall at 2 m/s. How fast is the top sliding down when the bottom is 6 m from the wall?
 $x^2 + y^2 = 100$. Differentiating: $2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$. At $x = 6$, $y = 8$: $12(2) + 16 \frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -1.5 \text{ m/s}$ (the top is sliding down at 1.5 m/s).

Volume and other rates

- 3 Air is pumped into a spherical balloon at $100 \text{ cm}^3/\text{s}$. Find the rate of change of the radius when $r = 5 \text{ cm}$, to three decimal places.
 $V = \frac{4}{3}\pi r^3$, so $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. $100 = 4\pi(25) \frac{dr}{dt}$, so $\frac{dr}{dt} = \frac{1}{\pi} \approx 0.318 \text{ cm/s}$.
- 4 Water drains from a conical tank (vertex down) with radius 4 m and height 10 m at the top, at a rate of $2 \text{ m}^3/\text{min}$. Using similar triangles $r = \frac{2h}{5}$, find how fast the water level is dropping when $h = 5 \text{ m}$.
 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75} h^3$. So $\frac{dV}{dt} = \frac{12\pi}{75} h^2 \frac{dh}{dt}$. At $h = 5$: $-2 = \frac{12\pi}{75}(25) \frac{dh}{dt}$, giving
 $\frac{dh}{dt} = -\frac{2 \times 75}{300\pi} = -\frac{1}{2\pi} \approx -0.159 \text{ m/min}$.

Angle and distance rates

- 5 A camera tracks a rocket launching vertically, 500 m from the launch pad. When the rocket is 1200 m high and rising at 200 m/s, how fast is the camera's angle of elevation increasing? Give the answer in radians/s, to four decimal places.
 $\tan \theta = \frac{h}{500}$, so $\sec^2 \theta \frac{d\theta}{dt} = \frac{1}{500} \frac{dh}{dt}$. At $h = 1200$: hypotenuse $= \sqrt{500^2 + 1200^2} = 1300$,
 $\sec^2 \theta = \left(\frac{1300}{500}\right)^2 = 6.76$. $6.76 \frac{d\theta}{dt} = \frac{200}{500} = 0.4$, so $\frac{d\theta}{dt} \approx 0.0592 \text{ rad/s}$.
- 6 Two cars leave an intersection at the same time, one heading north at 40 mph and the other heading east at 30 mph. How fast is the distance between them increasing after 2 hours?
 $x = 60 \text{ mi}$, $y = 80 \text{ mi}$ (east, north after 2 h). $s^2 = x^2 + y^2$: $2s \frac{ds}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$. $s = 100$:
 $100 \frac{ds}{dt} = 60(30) + 80(40) = 1800 + 3200 = 5000$, so $\frac{ds}{dt} = 50 \text{ mph}$.

More related rates

- 7 A streetlight is 6 m high. A 1.8 m tall person walks away from it at 1.5 m/s. Using similar triangles, find how fast the length of the person's shadow is growing.

Let x = distance from pole, s = shadow length. Similar triangles: $\frac{6}{x+s} = \frac{1.8}{s}$, so $6s = 1.8x + 1.8s$, giving $4.2s = 1.8x$, $s = \frac{1.8}{4.2}x = \frac{3}{7}x$. $\frac{ds}{dt} = \frac{3}{7} \cdot 1.5 \approx 0.643$ m/s.

- 8 The volume of a cube is increasing at $12 \text{ cm}^3/\text{s}$. Find the rate of change of the side length when the side is 3 cm.

$V = s^3$, $\frac{dV}{dt} = 3s^2 \frac{ds}{dt}$. $12 = 3(9) \frac{ds}{dt}$, so $\frac{ds}{dt} = \frac{12}{27} = \frac{4}{9}$ cm/s.

Rates involving trigonometric relationships

- 9 A kite is flying at a constant height of 40 m, moving horizontally away from the person holding the string at 3 m/s. Find the rate at which the string length is increasing when the kite is 30 m horizontally from the person.

$L^2 = 40^2 + x^2$: $2L \frac{dL}{dt} = 2x \frac{dx}{dt}$. At $x = 30$: $L = \sqrt{1600 + 900} = 50$. $50 \frac{dL}{dt} = 30(3) = 90$, so $\frac{dL}{dt} = 1.8$ m/s.

- 10 A conical pile of sand has its height always equal to its base radius. Sand is added at $10 \text{ m}^3/\text{min}$. Find the rate of change of the height when $h = 5$ m, using $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3$.

$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$. At $h = 5$: $10 = \pi(25) \frac{dh}{dt}$, so $\frac{dh}{dt} = \frac{10}{25\pi} = \frac{2}{5\pi} \approx 0.127$ m/min.